

Jacob Tenorio

Mechanical Engineering B.S. focusing on Aerospace Engineering

Washington University in St. Louis

About Me

I am the Lead Structures & Test Engineer for WashU Design/Build/Fly. Through this club, I have cultivated an interest in aeronautical engineering. Through coursework in fluid mechanics, heat transfer, and thermal systems design, I have developed a parallel interest in thermal-fluid sciences. My goal is a career in a discipline such as turbomachinery, hot-section engineering, environmental control systems, or another branch of thermal management.

This interest dates back to my youth, when I helped my father with HVAC installations on weekend jobs. As I grew older, I began to wonder how air-conditioning units actually worked, which drew me toward power transfer; what I later learned in university was really thermal energy transfer. This intersection of interest in aircraft and thermal-fluid sciences has shaped an engineering career path I am eager to pursue.

As the Structures and Test lead, I oversee smaller sub-teams of students, each responsible for components critical to that year's competition missions. This has included addressing challenges like transportation constraints, which led one aircraft to feature removable wings. After a crash at a competition event, I expanded what had been a purely Structures team into Structures & Testing, introducing a greater emphasis on real-world validation than the team had previously used. This took the form of physical test rigs, used both to validate simulation results and to run niche experiments that are difficult to simulate. Competing under limited resources, especially in the aftermath of a crash, has sharpened my ability to think on my feet, identify weaknesses in existing methodologies, and improve the effectiveness of the teams I am part of and go on to lead. I weigh every idea my team brings to me, and I encourage my teammates to generate plenty of ideas, even ones that go unused, since the strongest solution tends to emerge from having many to choose from.

Education

Washington University in St. Louis

Expected May 2027

B.S. in Mechanical Engineering **GPA: 3.51**

Relevant Coursework: Fluid Mechanics, Solid Mechanics, Heat Transfer, Thermal Systems Design, Thermodynamics, Machine Elements, Numerical Methods

Jacob Tenorio

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EDUCATION

Washington University in St. Louis *Degree Expected May 2027*
B.S. in Mechanical Engineering | Germany Ballintyn Scholar | GPA: 3.51 | Dean's List: Fall 2023, Spring 2025, Fall 2025
Relevant Coursework: Thermal Systems Design, Aircraft Design, Fluid & Solid Mechanics, Heat Transfer, Materials Selection

PROJECTS

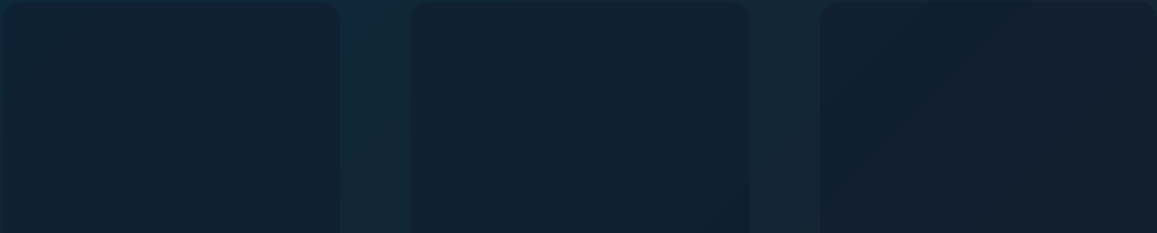
AIAA Design/Build/Fly & SAE Aero Design Competition Airplanes *August 2023 – Present*

- Lead 15-member structures and testing team to design competition-compliant aircraft via SolidWorks top-down modeling
- Design mission-specific components (e.g., deployable glider, towable sensor, removable wings, pylons, cargo compartments)
- Validate structural and aerodynamic components (e.g., bulkheads, winglets, spar lengths) via CFD, FEA, and test rigs

Projects

Multi-Mission Competition Aircraft

AIAA DESIGN/BUILD/FLY **SAE AERO DESIGN** | **SOLIDWORKS** **ANSYS** **CFD** **FEA**



WHAT?

- Designed aircraft for deployment missions (gliders/banners) and heavy cargo (passengers/pylons)
- Balanced high-lift capacity with low structural weight
- Lead structures and testing teams designing competition-compliant aircraft via SolidWorks top-down modeling

HOW?

- Modeled aircraft in **SolidWorks** using top-down modeling
- Integrated carbon fiber (spars), balsa (ribs), **3D-printed** PLA (complex mounts), and **CNC** machined parts (wingbox)
- Validated structural integrity (**FEA**) and aerodynamic performance (**CFD**)
- Designed mission-specific components (removable wings, deployable glider, cargo & passenger compartments)

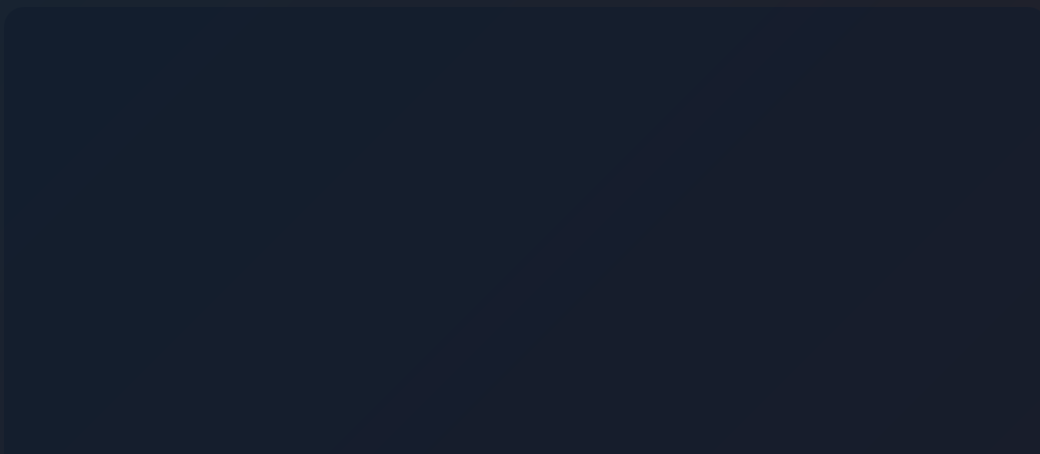
RESULTS

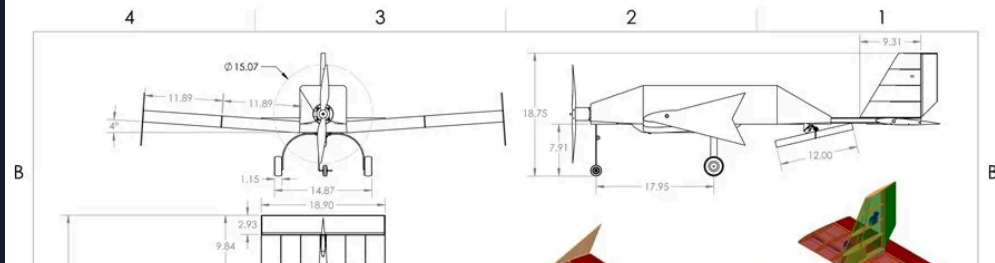
- Placed **35th of 114** in international AIAA competition
- Achieved **100%** structural success rate with no components failing during flight
- Reduced new-member CAD training time by **50%** by designing an onboarding curriculum

DESIGN REPORTS

Primary Author (2026), Contributing Author (2025) – Detail Design Section

CAD DRAWINGS FROM 2026 REPORT





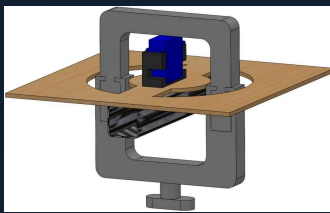
Glider Deployment Mechanism

AEROSPACE

SOLIDWORKS

FEA

3D PRINTING



WHAT?

- Designed lightweight release mechanism for mid-flight glider deployment
- Optimized for small-scale geometry and minimal weight impact on parent aircraft

HOW?

- Developed multiple **SolidWorks** designs, utilizing **3D-printed** PLA to test and refine release efficiency
- Engineered a one-step release mechanism to eliminate mechanical **failure points** and ensure mission reliability
- Performed stress analysis via SolidWorks **FEA** to identify critical points

RESULTS

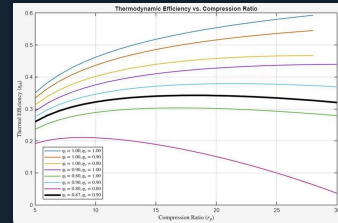
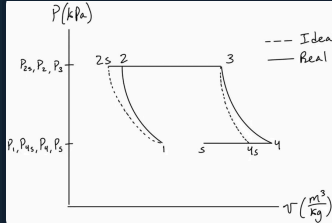
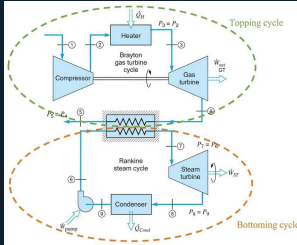
- Achieved **jam-free release** during Design/Build/Fly competition
- Simplified design proved effective under real-world flight vibrations and loads

CCGT Marine Propulsion System Design

THERMAL SYSTEMS

MATLAB

PYTHON



WHAT?

- Designed a combined-cycle gas turbine marine propulsion system
- Optimized both Brayton and Rankine cycles for maximum efficiency

HOW?

- Optimized variable-specific-heat Brayton cycle in **MATLAB** via parametric compression ratio and isentropic efficiency sweeps
- Maximized Rankine efficiency in **Python** via turbine pressure (optimization with blade-erosion quality constraint
- Sized three-stage HRSG via Dittus-Boelter, Zukauskas, Rohsenow correlations, LMTD, and thermal resistance networks

RESULTS

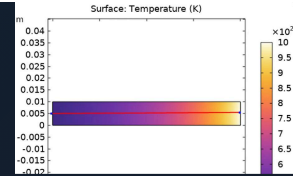
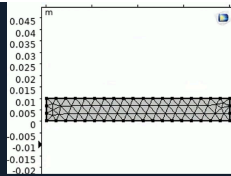
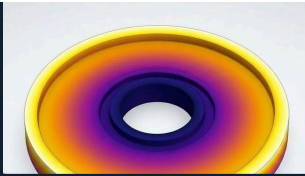
- Delivered optimized combined-cycle system meeting all design constraints
- Balanced thermodynamic efficiency with practical engineering constraints (blade erosion, quality limits)

Radial Heat Conduction in a Micro-Turbine Disk

HEAT TRANSFER

COMSOL

FEA



WHAT?

- Simulated steady-state radial conduction in a micro-turbine disk
- Evaluated superalloy materials for turbine shaft thermal protection

HOW?

- Built **COMSOL** axisymmetric model with fixed rim temperature, forced convection, and lubricant cooling BCs
- Validated against Bessel fin analytical solution
- Performed parametric study and mesh independence testing across 3 superalloys

RESULTS

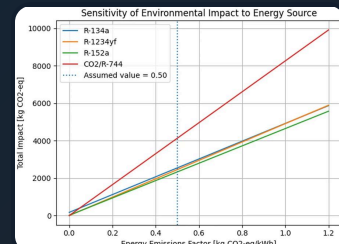
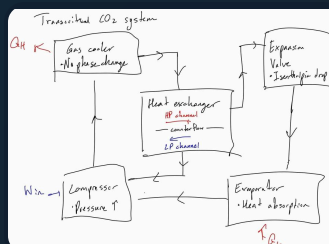
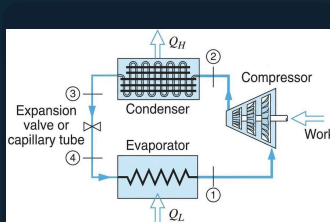
- Confirmed **<6%** deviation between COMSOL and analytical solution
- Identified **René 41** as optimal material for shaft thermal protection
- Demonstrated mesh-independent convergence

Automotive HVAC Refrigerant Cycle Simulation

THERMODYNAMICS

PYTHON

TEWI ANALYSIS



WHAT?

- Modeled four automotive refrigerants across conventional and transcritical vapor-compression cycles in Python
- Investigated lifecycle environmental impact of refrigerant selection

HOW?

- Built simulation in **Python** covering conventional and transcritical vapor-compression cycles
- Optimized CO₂/R-744 transcritical high-side pressure to maximize COP
- Balanced cooling capacity against compressor work

RESULTS

- Conducted TEWI analysis revealing **indirect emissions dominate** climate impact
- Found near-zero GWP insufficient to minimize total environmental footprint

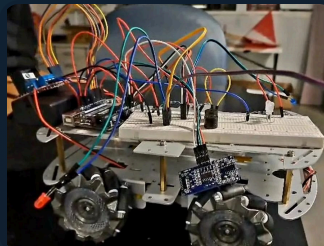
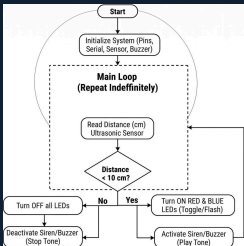
Autonomous Ambulance

ROBOTICS

ARDUINO

C++

SOLIDWORKS



WHAT?

- Developed autonomous driving system with collision avoidance, adaptive siren logic, and patient transport ability
- Focused on reducing vehicle weight and project overhead

HOW?

- Modeled chassis in **SolidWorks**
- Developed **C++** control firmware for **Arduino** microcontroller and implemented associated circuitry
- Integrated **ultrasonic (sonar) sensors** for autonomous navigation and obstacle avoidance
- Implemented functional emergency lights and sirens

RESULTS

- Achieved full vehicle autonomy
- Demonstrated **100%** obstacle avoidance success
- Validated patient transport capabilities
- Reduced chassis weight by **50%** via design optimization
- Completed project **30%** under budget

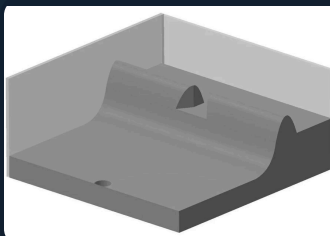
Flood Disaster Model

ARMY CORPS OF ENGINEERS

CFD

SOLIDWORKS

3D PRINTING



WHAT?

- Designed model to simulate water propagation following a catastrophic levee failure
- Focused on creating model that accurately replicates fluid behavior

HOW?

- Used **SolidWorks CFD** to optimize design for maximum flow realism
- Sliced and printed final design via **3DprinterOS**

RESULTS

- Produced **3D-printed PETG** prototype for physical validation and testing
- Delivered validated design and fabrication-ready files to subsequent development team

Skills

Technical

SolidWorks

Ansys

MATLAB

Python

Java

COMSOL

XFLR5

Arduino

Excel

Methods

Computational Fluid
Dynamics

Finite Element
Analysis

AI-Assisted
Workflows

GD&T

Equipment

CNC

Lathe

Mill

Drill Press

Laser Cutter

3D Printing

Band Saw

Soldering

Brazing

Get in Touch

Interested in connecting? I'd love to hear from you.



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